

INPUT

```
/**
```

```
* Write a description of class Vehicle here.
```

```
*
```

```
* @author (BITA/7/23/006/TZ)
```

```
* @version (22/01/2026)
```

```
*/
```

```
class Vehicle
```

```
{
```

```
    protected String brand;
```

```
    protected String model;
```

```
    protected int year;
```

```
    protected double price;
```

```
    public Vehicle(String brand, String model, int year, double price){
```

```
        this.brand=brand;
```

```
        this.model=model;
```

```
        this.price=price;
```

```
        System.out.println("Vehicle constructor called");
```

```
    }
```

```
    public void displayInfo(){
```

```
        System.out.println("Brand of vehicle:" + brand);
```

```
        System.out.println("Model of vehicle:" + model);
```

```
        System.out.println("year of vehicle:" + year);
```

```
        System.out.println("price of vehicle:" + price);
```

```

    }

    public void startEngine(){
        System.out.println(brand+" "+ "Engine started");
    }

    public void stopEngine(){
        System.out.println(brand + " " +"engine stopped");
    }
}

class Car extends Vehicle{
    private int numDoors;
    private String fuelType;

    public Car(String brand, String model,int year, double price, int numDoors,String fuelType){
        super(brand,model,year,price);
        this.numDoors=numDoors;
        this.fuelType=fuelType;
        System.out.println("Car consructor called");

    }

    @Override
    public void displayInfo(){
        super.displayInfo();

        System.out.println("Doors:"+numDoors);
        System.out.println("Fuel Type:" +fuelType);

    }
}

```

```

public void honk()
{
    System.out.println(brand+ " " +model+ "says: Beep! Beep!");
}

public void openTrunk(){
    System.out.println("Opening trunk of :"+ brand +"" +model);
}
}

class motorcycle extends Vehicle {
    private int engineCC;
    private boolean hasHelmentStorage;

    public motorcycle(String brand, String model,int year,double price, int engineCC, boolean
hasHelmentStorage){
        super( brand,model,year,price);
        this.engineCC=engineCC;
        this.hasHelmentStorage=hasHelmentStorage;
        System.out.println("Motorcycle constructor called");
    }

    @Override
    public void displayInfo(){
        super.displayInfo();
        System.out.println("Engine:" +engineCC+ "CC");
        System.out.println("Helmet Storage:" +(hasHelmentStorage? "Yes": "No"));
    }
}

```

```

    public void wheelie(){
        System.out.println(brand+ "" +model+ "is doing a wheelie!");
    }
    public void kickStart(){
        System.out.println("Kick starting" +brand+ " "+model);
    }
}

class ElectricCar extends Car{
    private double batteryCapacity;
    private int range;

    public ElectricCar(String brand,String model, int year,double price,int numDoors,String
fuelType,double batteryCapacity,int range){
        super( brand,model,year,price,numDoors,fuelType);
        this.batteryCapacity=batteryCapacity;
        this.range=range;
        System.out.println("Electricar constructor called");
    }
    @Override
    public void startEngine(){
        System.out.println(brand+ "" +model+"electric motor activated silently!");
    }
    public void charge(){
        System.out.println("charging" +brand+"" +model+"....battery;" + batteryCapacity+"km");
    }
}

```

```

public void displayBatteryStatus(){
    System.out.println("battery capacity:"+batteryCapacity+"km");
    System.out.println("range: "+range+"km");
}
}

public class excercise_12Inheritance{
public static void main(String[] args){
    System.out.println("===VEHICLE HIERACHY TEST ===\n");
    Car car= new Car("toyota","corolla",2023,45000000,4,"petrol");
    motorcycle motorcycle=new motorcycle("honda","CBR500R",2022,1500000,500,false);
    ElectricCar ElectricCar=new
ElectricCar("tesla","model3",2024,85000000,4,"electric",75.0,450);

    System.out.println("---testing car---");
    car.displayInfo();
    car.startEngine();
    car.honk();
    car.openTrunk();
    car.stopEngine();

    System.out.println("\n---testing motorcyle---");
    motorcycle.displayInfo();
    motorcycle.startEngine();
    motorcycle.wheelie();
    motorcycle.kickStart();
    motorcycle.stopEngine();
}
}

```

```
System.out.println("\n---testing Electric Car---");

ElectricCar.displayInfo();

ElectricCar.startEngine();

ElectricCar.charge();

ElectricCar.displayBatteryStatus();

ElectricCar.honk();

ElectricCar.stopEngine();


System.out.println("\n---polymorphism demo---");

Vehicle[] vehicles={car,motorcycle,ElectricCar};

for (Vehicle v :vehicles){

    v.displayInfo();

    v.displayInfo();

    System.out.println();

}

System.out.println("== END OF TEST ===");

}

}
```

OUTPUT

Options

```
===VEHICLE HIERACHY TEST ===  
  
Vehicle constructor called  
Car consructor called  
Vehicle constructor called  
Motorcycle constructor called  
Vehicle constructor called  
Car consructor called  
Electricar constructor called  
---testing car---  
Brand of vehicle:toyota  
Model of vehicle:corolla  
year of vehicle:0  
price of vehicle:4.5E7  
Doors:4  
Fuel Type:petrol  
toyota Engine started  
toyota corollasays: Beep! Beep!  
Opening trunk of :toyotacorolla  
toyota engine stopped  
  
---testing motorcyle---  
Brand of vehicle:honda  
Model of vehicle:CBR500R  
year of vehicle:0  
price of vehicle:1500000.0  
Engine:500CC
```

Can only enter input while your program is running

---testing motorcyle---

Brand of vehicle:honda

Model of vehicle:CBR500R

year of vehicle:0

price of vehicle:1500000.0

Engine:500CC

Helmet Storage:No

honda Engine started

hondaCBR500Ris doing a wheelie!

Kick startinghonda CBR500R

honda engine stopped

---testing Electric Car---

Brand of vehicle:tesla

Model of vehicle:model3

year of vehicle:0

price of vehicle:8.5E7

Doors:4

Fuel Type:electric

teslamodel3electric motor activated silently!

chargingteslamodel3....battery;75.0km

battery capacity:75.0km

range: 450km

tesla model3says: Beep! Beep!

tesla engine stopped

Options

```
---polymorphism demo---  
Brand of vehicle:toyota  
Model of vehicle:corolla  
year of vehicle:0  
price of vehicle:4.5E7  
Doors:4  
Fuel Type:petrol  
Brand of vehicle:toyota  
Model of vehicle:corolla  
year of vehicle:0  
price of vehicle:4.5E7  
Doors:4  
Fuel Type:petrol  
  
Brand of vehicle:honda  
Model of vehicle:CBR500R  
year of vehicle:0  
price of vehicle:1500000.0  
Engine:500CC  
Helmet Storage:No  
Brand of vehicle:honda  
Model of vehicle:CBR500R  
year of vehicle:0  
price of vehicle:1500000.0  
Engine:500CC  
Helmet Storage:No
```

Brand of vehicle:tesla
Model of vehicle:model3
year of vehicle:0
price of vehicle:8.5E7
Doors:4
Fuel Type:electric
Brand of vehicle:tesla
Model of vehicle:model3
year of vehicle:0
price of vehicle:8.5E7
Doors:4
Fuel Type:electric

== END OF TEST ==

